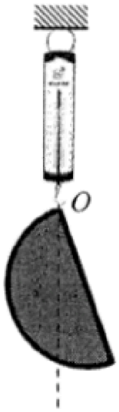
2013 Q14

14.



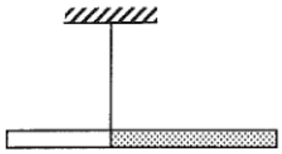
A semi-circular cardboard hangs from a spring balance from point O as shown. The reading of the spring balance is 5 N. Which statements are correct ?

(1) The weight of the cardboard is 5 N.
(2) The centre of gravity of the cardboard is directly under O.
(3) The reading of the balance would become zero if the set-up is brought to the Moon's surface.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2014 Q3

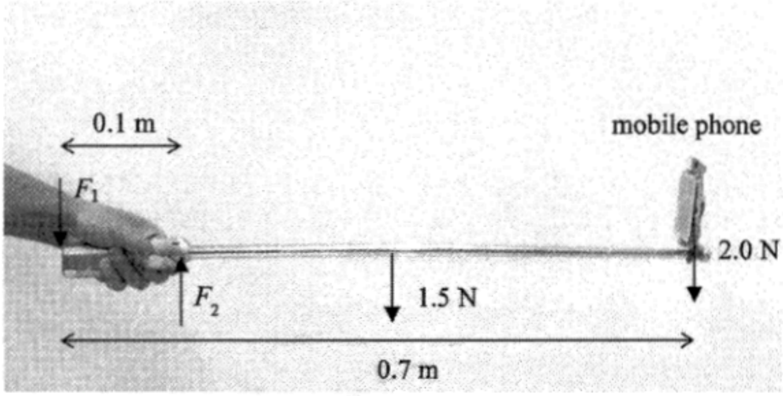
3.



PQR is a composite rod having a uniform cross-section but with portions PQ and QR made of different materials, each of uniform density. The ratio of the length of PQ to that of QR is 2 : 3. When the rod is suspended at Q, it remains horizontal as shown. What is the ratio of the mass of PQ to that of QR ?

A. 2 : 3
B. 1 : 1
C. 3 : 2
D. Answer cannot be found as the ratio of their densities is not given.

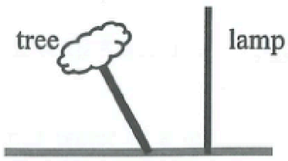
8. Selfie sticks are popular nowadays. A uniform selfie stick of length 0.7 m is held horizontally as shown. Assume that the forces required to hold the selfie stick by the hand are represented by F_1 and F_2, and F_1 and F_2 are perpendicular to the stick.



It is given that the weight of the selfie stick and the mobile phone are 1.5 N and 2.0 N respectively. Taking the mobile phone as a point mass, estimate the magnitude of F_2.

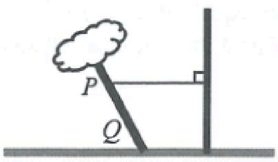
A. 3.5 N
B. 19.3 N
C. 35 N
D. Cannot be determined as F_1 is unknown.

7.

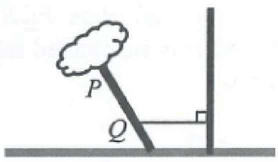


A tree blown by strong wind is found leaning to one side. In order to give the tree some support, a rope is wrapped around its trunk and tied to a fixed lamp post nearby. In which of the following arrangements would the rope have the highest chance of breaking ?

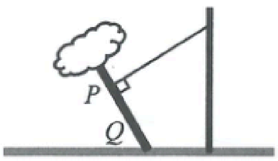
A.



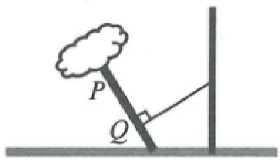
B.



C.

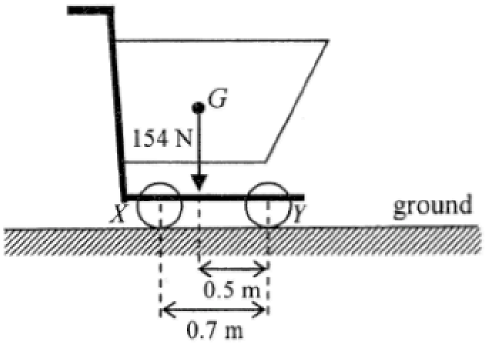


D.



2013 Q9

9.



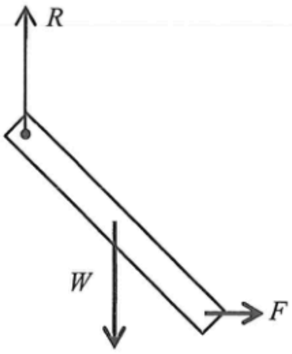
The figure shows a supermarket trolley resting on the ground. The separation between cylindrical wheels X and Y is 0.7 m. When the trolley is loaded to a total weight of 154 N, its centre of gravity G is at a horizontal distance of 0.5 m from the wheel Y. What is the reaction acting on the wheel X from the ground ?

A. 44 N
B. 62 N
C. 92 N
D. 110 N

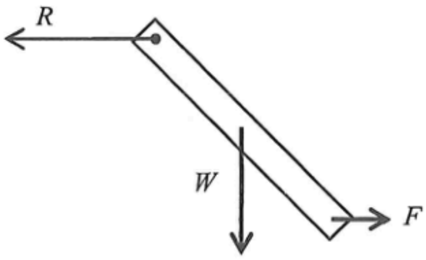
2025 Q7

7. A uniform rod of weight W hinged at its upper end is at rest under a horizontal force F applied at its lower end. Which figure correctly shows the force R acting on the rod at the hinge ?

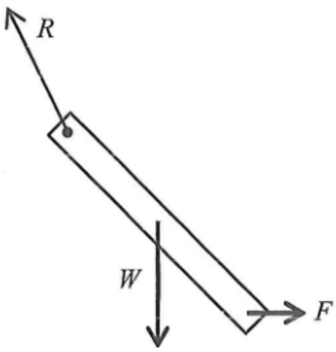
A.



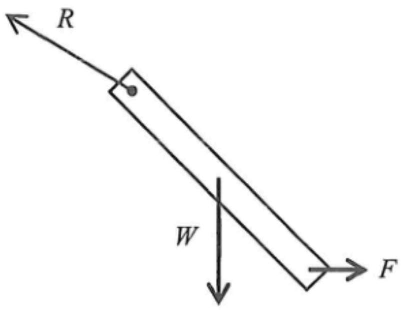
B.



C.

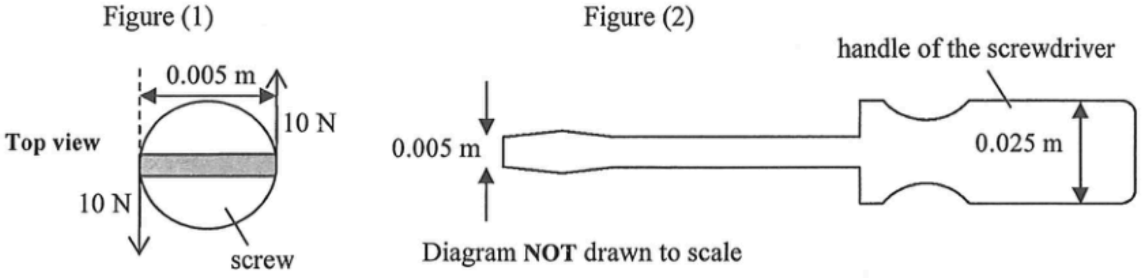


D.



2025 Q10

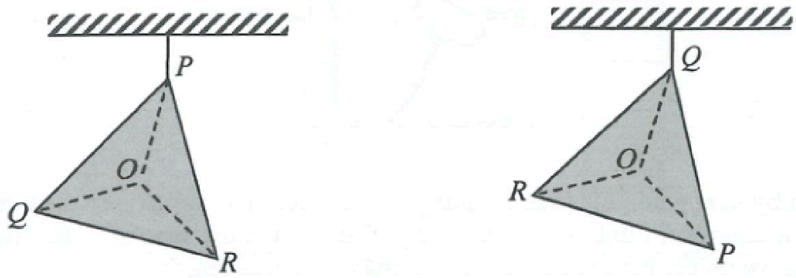
10. Loosening a screw requires a pair of forces of 10 N acting tangentially as shown in Figure (1). A suitable screwdriver with a handle of diameter 0.025 m shown in Figure (2) is chosen. Estimate the magnitude of the pair of forces needed to act tangentially on the handle of the screwdriver.



A. 2 N
B. 4 N
C. 5 N
D. 10 N

2019 Q4

4.

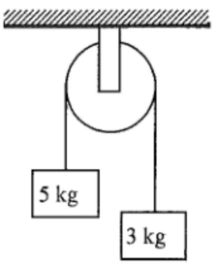


O is the centre of a metal plate PQR in the form of an equilateral triangle with **non-uniform** mass distribution. The plate is suspended from the ceiling at P and then at Q as shown. The centre of gravity of the metal plate is

A. at O.
B. within the region POQ.
C. within the region ROQ.
D. within the region POR.

2014 Q8

8. Two blocks of masses 5 kg and 3 kg respectively are connected by a light string passing through a frictionless fixed light pulley. Find the magnitude of the acceleration of the blocks in terms of the acceleration due to gravity g when they are released. Neglect air resistance.

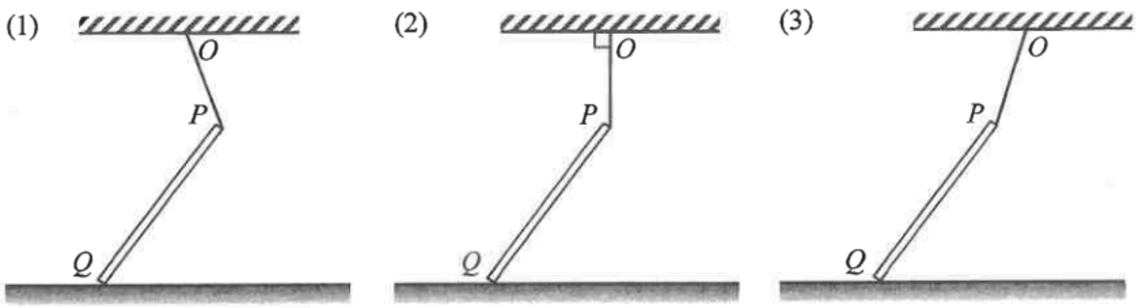


A. g

B. $\frac{g}{\ }$

2022 Q5

5. In each figure below, a uniform rod PQ has its upper end P attached to point O on the ceiling by a light inextensible string. The rod's lower end Q rests on a rough horizontal ground as shown.

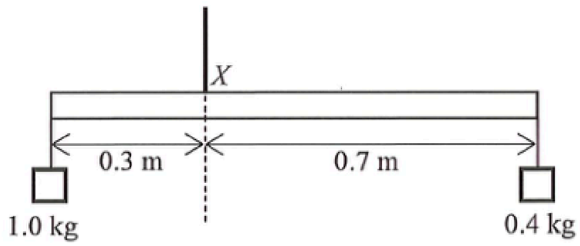


In which figure is the frictional force acting on the rod due to the ground towards the right ?

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

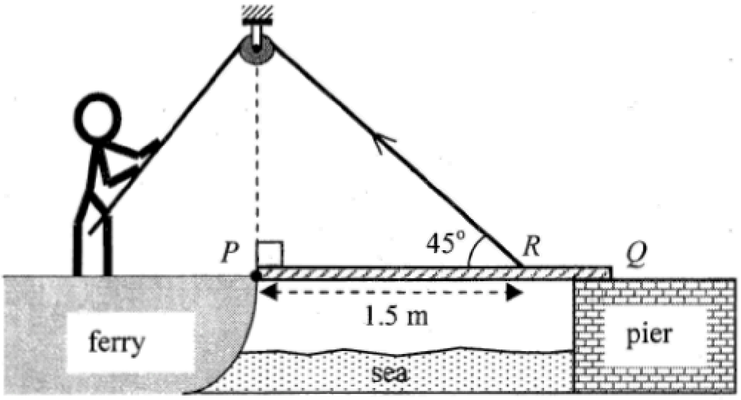
2023 Q12

12. A uniform metre rule of a certain weight is hung by a string at X such that it is kept in equilibrium with weights hanging at both ends as shown. If the 1.0-kg weight is shifted 0.1 m towards X, what distance should the 0.4-kg weight be shifted towards X in order to restore equilibrium ?



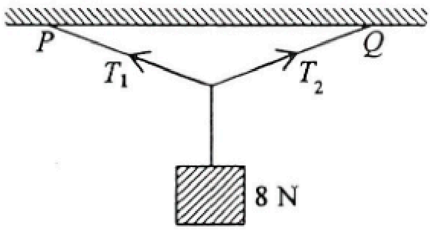
A. 0.1 m
B. 0.2 m
C. 0.25 m
D. 0.45 m

6. A uniform gangplank PQ of a ferry smoothly hinged at end P initially rests horizontally on the pier. The gangplank has mass M and length 2 m. It is raised by a man on the ferry using a light rope passing a smooth fixed light pulley and connecting to R on the gangplank as shown. R is 1.5 m from end P. Which of the following correctly describes the force required to raise the gangplank steadily ?



	initial force required to raise the gangplank when it is horizontal	*subsequent* force required to raise the gangplank
A.	0.67 Mg	greater than 0.67 Mg
B.	0.67 Mg	smaller than 0.67 Mg
C.	0.94 Mg	greater than 0.94 Mg
D.	0.94 Mg	smaller than 0.94 Mg

5. A block of weight 8 N is suspended from a horizontal ceiling by light inextensible strings to tw different points P and Q as shown. The strings are equal in length.



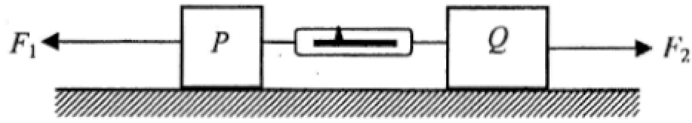
Which of the following descriptions about the tensions T_1 and T_2 in the two strings is/are correct ?

(1) The magnitude of T_1 must be greater than 4 N.
(2) The maximum value of T_2 would not exceed 8 N.
(3) The resultant force of T_1 and T_2 is zero.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2012 Q8

8.



Blocks P and Q of mass m and $2m$ respectively are connected by a light spring balance and placed on a smooth horizontal surface as shown. If horizontal forces F_1 and F_2 (with $F_1 > F_2$) act on P and Q respectively and the whole system moves to the left with constant acceleration, what is the reading of the spring balance ?

A. $\dfrac{2F_1 - F_2}{3}$

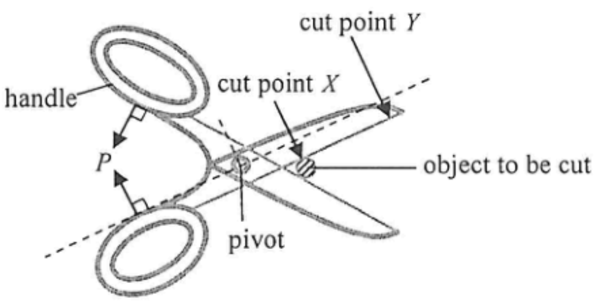
B. $\dfrac{2(F_1 - F_2)}{3}$

C. $\dfrac{2F_1 + F_2}{3}$

D. $\dfrac{F_1 + 2F_2}{3}$

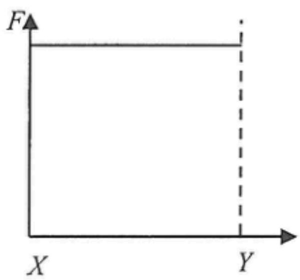
2020 Q7

7. The figure shows that a pair of forces P of constant magnitude is applied at right angles to the handles of a pair of scissors in order to cut an object.

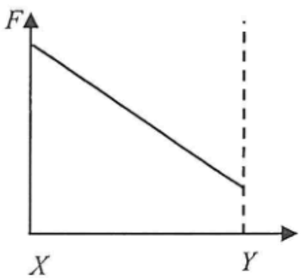


Which graph below best shows the variation of force F produced at cut point from X to Y when the pair of scissors is closed ?

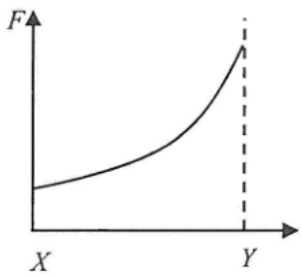
A.



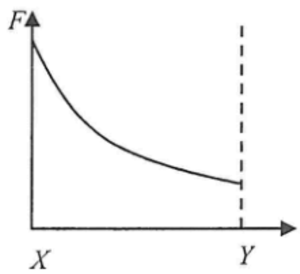
B.



C.

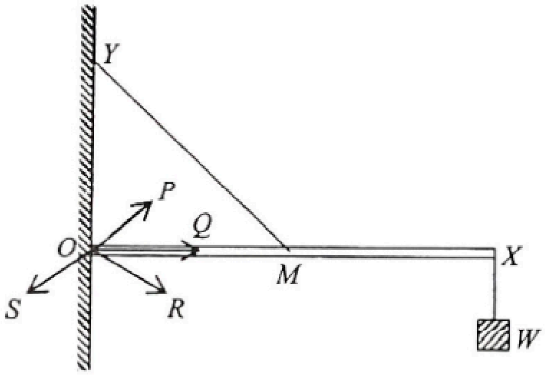


D.



2018 Q6

6.

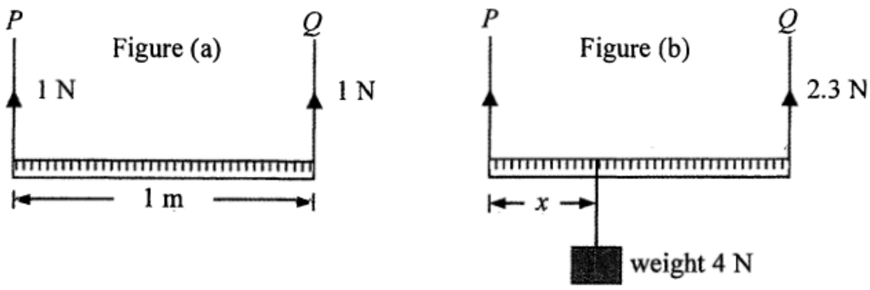


A uniform light rigid rod OX is hinged smoothly to a wall at one end O. Its mid-point M is connected by a light inextensible string to a point Y directly above O while a weight W is suspended from the other end X of the rod as shown. Rod OX remains horizontal. The reaction force acting on the rod due to the wall is along the direction

A. OP.
B. OQ.
C. OR.
D. OS.

2016 Q11

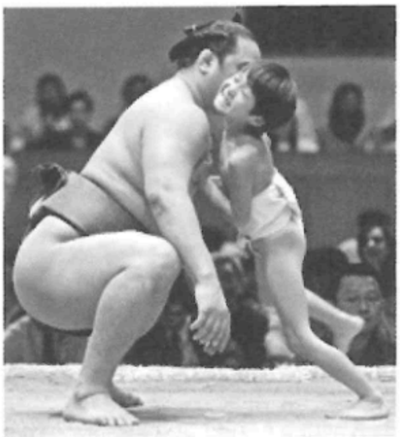
11.

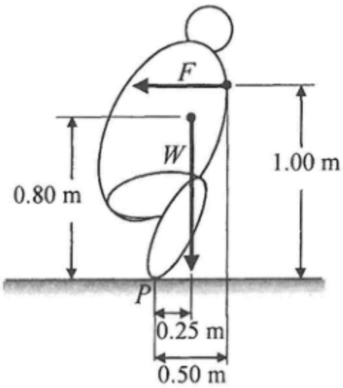


A uniform metre rule is supported by vertical wires P and Q and remains at rest horizontally as shown in Figure (a). The tension in each wire is 1 N. When a weight of 4 N is hung from the metre rule at a certain position as shown in Figure (b), the tension in Q becomes 2.3 N while the metre rule remains horizontal. Find the distance x shown.

A. 32.5 cm
B. 57.5 cm
C. 67.5 cm
D. Answer cannot be found as the tension in P is not known.

10. A child tries to push a 150 kg (sumo) wrestler in order to topple him backwards in a game. The simplified diagram indicates the weight W of the wrestler and the horizontal pushing force F acting on the wrestler by the child. P is the wrestler's contact point on the ground.

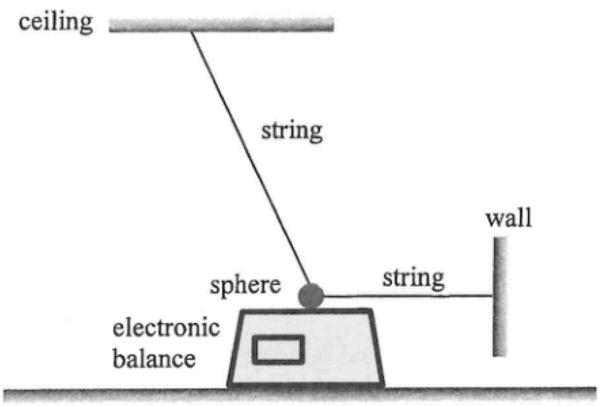




Find the minimum value of F required.

A. 368 N
B. 736 N
C. 1177 N
D. 2354 N

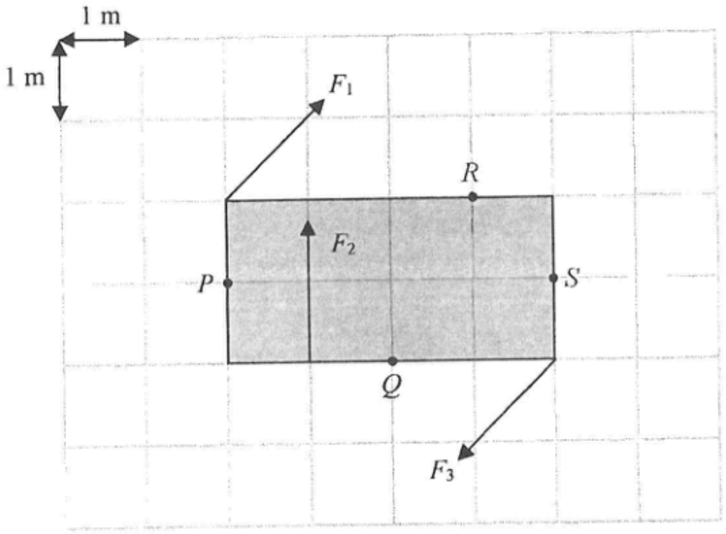
4.



A sphere of weight 18 N is suspended by two light and inextensible strings. One string is attached horizontally to a wall, and the other is connected to the ceiling. The sphere rests on the smooth top surface of an electronic balance, which shows a reading of 10 N. Find the resultant force acting on the sphere by the two strings.

A. 0 N
B. 8 N
C. 10 N
D. 15 N

7.



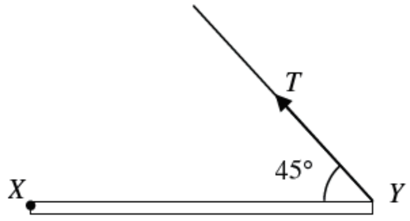
A rectangular plate of width 2 m and length 4 m is placed on a smooth horizontal surface. Three forces F_1, F_2 and F_3 of equal magnitude act on it as shown. F_1 and F_3 are parallel and opposite in direction. Taking the moment about which point below is the net moment the greatest ?

A. P
B. Q
C. R
D. S

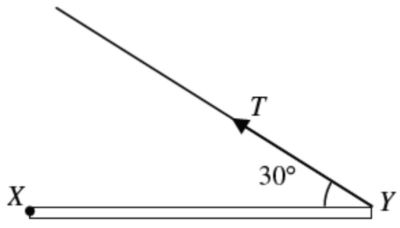
pp Q12

12. A rod XY hinged at X is kept horizontal by a light string. M is the midpoint of XY. In which of the following arrangements will the tension T in the string be the smallest ?

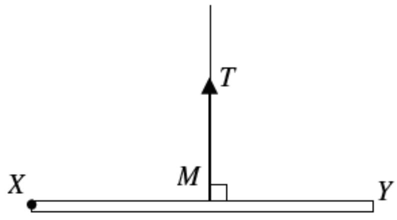
A.



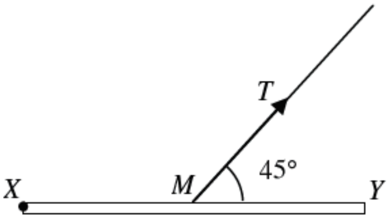
B.



C.



D.



10.

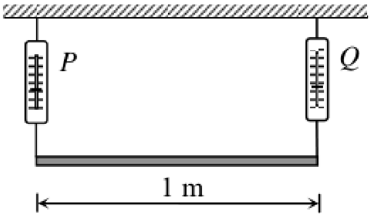


Figure (a)

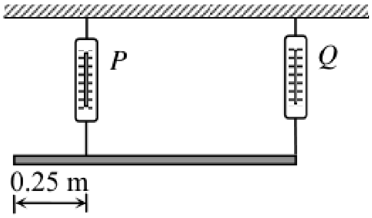


Figure (b)

Figure (a) shows a uniform plank supported by two spring balances *P* and *Q*. The readings of the two balances are both 150 N. *P* is now moved 0.25 m towards *Q* (see Figure (b)). Find the new readings of *P* and *Q*.

	Reading of *P*/N	**Reading of *Q*/N**
A.	100	200
B.	150	150
C.	200	100
D.	200	150